# FUNCTIONAL DESCRIPTION
# CONTAINER SERIES
## POWERGENERATION

For Type Series 4000

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Subject to technical alterations in the interest of further development.

# Table of Contents

# 1 General description

**The container unit is made up of the following main groups:**

- Steel construction container enclosure
- MTU Series 4000 (gas engine with generator mounted onto base frame)
- MMC (for system control, control, diagnosis)
- MIP (MTU Interface Panel) for engine control, regulating, and diagnosis
- Generator power control panel
- Gas system (incl. controlled gas system)
- Ventilation system
- Cooling system for engine cooling water and mixture cooling
- Heat extraction HP circuit (optional)
- Exhaust system
- Lube oil system

| NOTE | | |
|---|---|---|
| ! | The delivery object of MTU Onsite Energy satisfies the requirement specified in the EC Machinery Directive and in the German directives and standards. | SH-H-150 |

| NOTE | | |
|---|---|---|
| ! | When using the unit supplied in a foreign country or special regions, MTU Onsite Energy is not responsible for adherence to the legal or other regulations / standards at the place of use. | SH-H-151 |

| NOTE | | |
|---|---|---|
| ! | The images may not be identical to the actual design since some available customer options may be hidden or omitted on purpose. Furthermore, any tailor-made designs are not shown in these images. | SH-H-232 |

# 2 Container

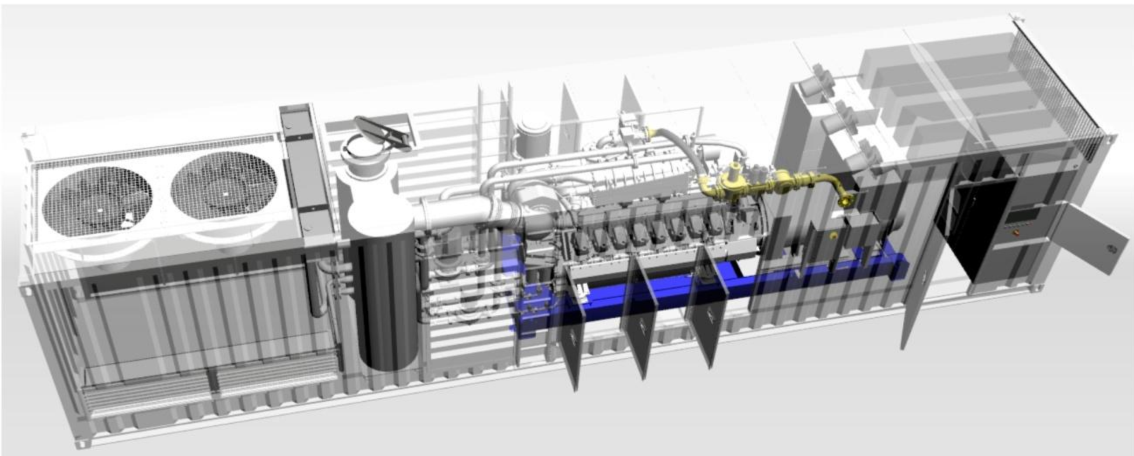
This container is a complete enclosure and comprises all necessary component necessary for the safe operation of the generators contained therein. This container unit has been thoroughly tested and is ready to use after connecting the necessary media at the client's site (fuel gas, power connections, third-party supply auxiliary drives, heat provided by the customer (if applicable). The tasks performed are limited to the installation of the roof structures (if applicable), connecting interfaces as described above, as well as those operating fluids available (cooling water, lubricants, etc.).

The container enclosure provides the necessary protection from the environment. The containment enclosure ensures that the noise created by the gas generators are kept to a predefined residual volume.

Access and maintenance doors are provided. Furthermore, certain structural elements are installed that make it possible to move components in and out of the container.

In order to protect the environment, the floor plates of the container are designed as an oil sump and equipped with detection devices (holding capacity 110 % of the fluids inside the container). This is compliant with the WHG (German Federal Water Act).

Depending on the series or their application, the technical data of the containers may change.



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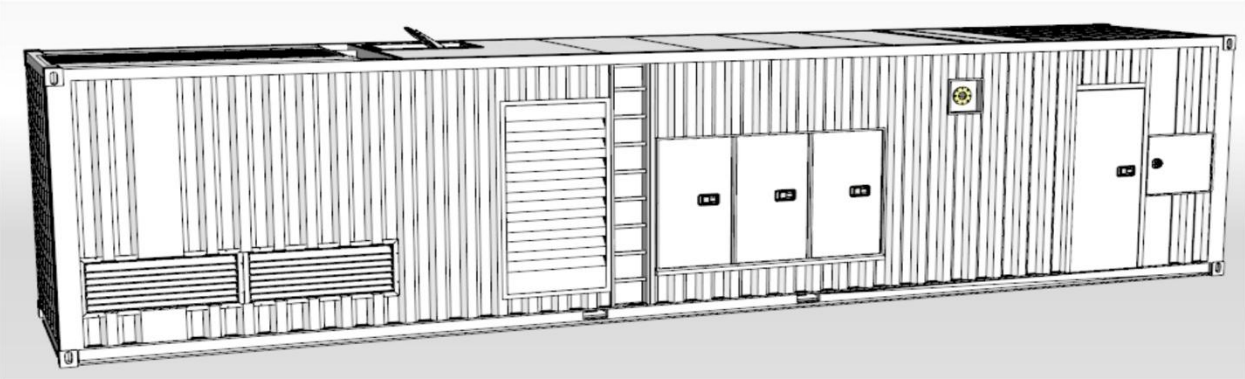
# 2.1 Series Powercontainer



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## 2.1.1 Container enclosure

A modified version of the ISO 40ft High Cube Container is the basis for this structure that will be used for special applications.
The container is stackable and easy to transport (CSC certified).

Basically, the container is divided into two rooms.
The machine room contains the generator, the control system, and the auxiliary drives. The radiator room is a separate area.
The generator is placed on the support rails that are anchored into the floor. All other components that are necessary for the operation of the generator, e.g., gas, lubrication oil, or cooling system as well as the control units and the generator's power control panel are located here.

The second partitioned area of the container houses the necessary heat exchangers for cooling the engine and mixture. Installation of the plate heat exchanger and optimizing the use of the generated heat.
The connections provided by the customer include gas, power supply, and auxiliary drives and are located along the side wall.
Connections for utilizing the heat (optional) of the engine cooling water are located on the container's roof.

## 2.1.2 Generator power control panel

The generator's power control panel is completely wired and mounted onto the side wall of the container.

**It comprises the following main components**

- Generator circuit breaker
- Connections provided by the customer
- Power supply for auxiliary drives
- Terminals and interfaces for customer communication

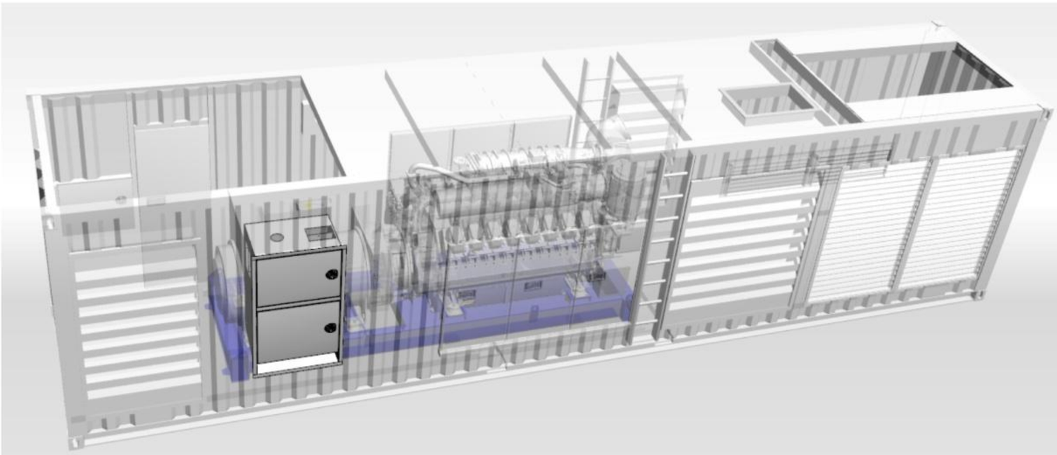
The generator power control panel is the interface to the customer's network.

This sub-assembly will also be used when disconnecting the system from the main power supply.

All necessary external feed-in lines for the most important axillary drives – which are necessary for the safe operation – are supplied via a separate feed-in point.

Accessibility/operation of the above-mentioned components from the outside.

Place of installation is along the side wall of the container.



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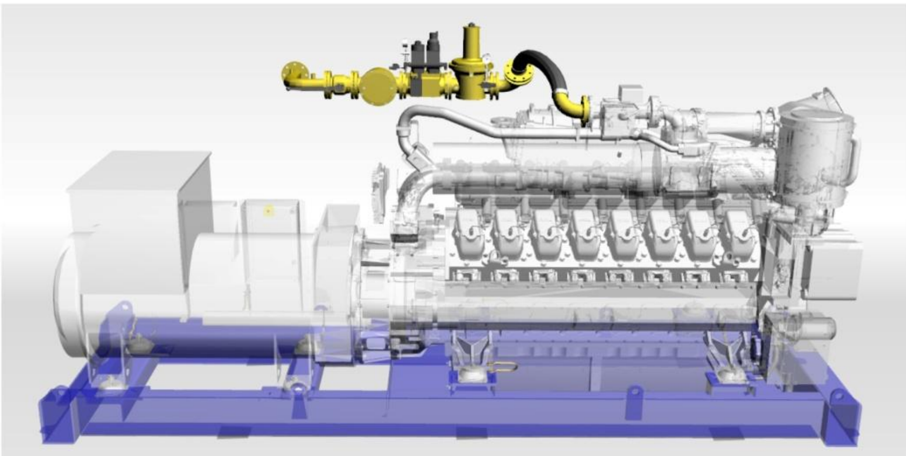
| NOTE | | |
|---|---|---|
| ! | Please refer to wire diagram or operating instructions. | SH-H-148 |

### 2.1.3 Gas system

**Fully assembly ready – gas control system:**
**It comprises the following components:**

- Gas filter
- Two solenoid valves (or a double solenoid valve)
- Low pressure regulator
- Valve tightness check
- Flexible stainless steel hose assembly
- Shut-off devices
- A safety shutoff valve is included for the installation into the gas line outside of the container.

A welded DIN-compliant flange in the container wall is used to provide an interface to the customer.



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| NOTE | | |
|---|---|---|
| ! | Please refer to operating instructions and manufacturer's documentation. | SH-H-148 |

| INFORMATION | | |
|---|---|---|
| | Compliance with the currently valid specifications for fluids and lubricants is mandatory! | SH-H-023 |

## 2.1.4 Ventilation system

**A ventilation system is integrated into the container and ensures the following features.**

- It ensures the proper amount of compressed air is supplied to the machine base (cp. safety concept).
- Transporting the engine's combustion air.
- Transporting the cooling air for the generator and other components installed in the system.

One or several fans are used to carry the necessary amount the air.

The operating mode of the fans as well as their activation may be indifferent, depending on the series of the container.

BR Powercontainer →Exhaust air fan

BR CHP Container→Supply air fan

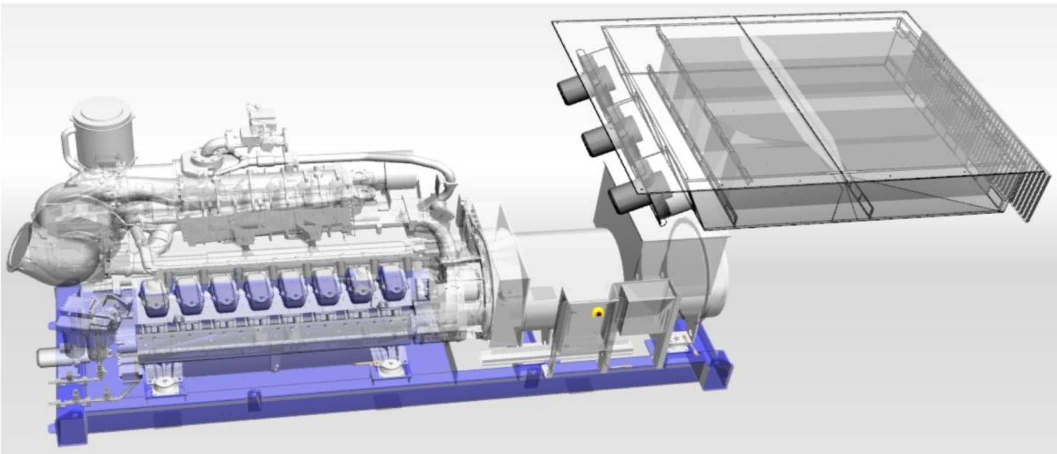
**The supply as well as the exhaust air ducts:**

**It comprises the following components:**

- Sound dampeners
- Weather protection and screens to keep out small animals

Outlet: Exhaust via front

Inlet: Supply air through the sidewalls



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| NOTE | | |
|---|---|---|
| ! | Refer to operating instructions | SH-H-148 |

## 2.1.5 Cooling system for engine cooling water and mixture cooling

For removal, the waste heat of the gas engine complete cooling systems are mounted.

**The cooling system consists of two separate circuits:**

Engine cooling water circuit (HT)

This removes the heat generated from the engine (engine block, oil cooler, etc.) and from the 1st stage of the mixture cooler.

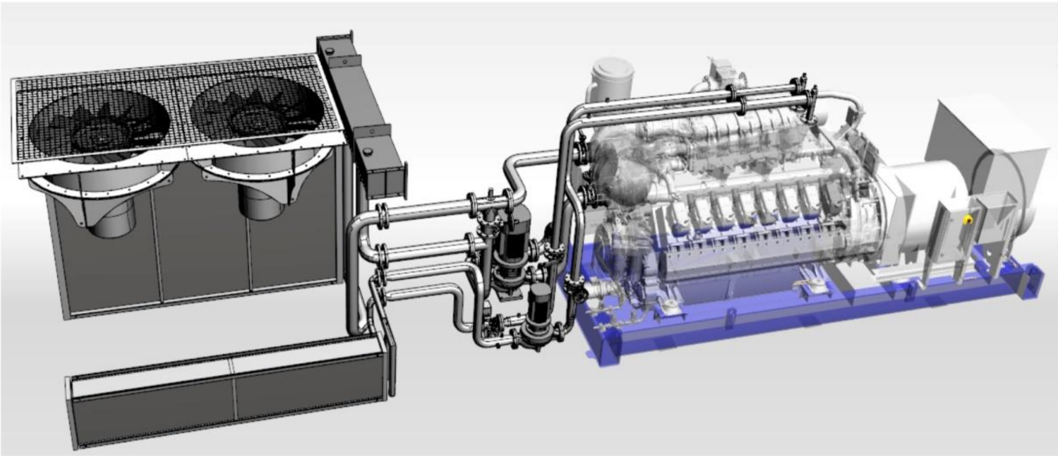
Mixture cooling circuit (LT)

This removes the heat generated from the 2nd stage of the mixture cooler.

The cooling system must be filled with a mixture of water and antifreeze/corrosion protection agent in accordance with the fluids and lubricants specifications of MTU Onsite Energy.

An electrical pump is used to deliver the cooling media. Mechanically or electrically operated mixing valves are used to control the necessary temperatures in the cooling system.

Radiators with electric fans are integrated into the container and ensure the heat exchange to the surroundings. The design of the radiators (sandwich/split core etc.) can vary.



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| NOTE | | |
|---|---|---|
| ! | Refer to operating instructions | SH-H-148 |